

**JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]**

ANSWER KEY

PAPER-1

**PART-1 : PHYSICS**

| SECTION-I  | Q. | 1     | 2     | 3     | 4              | 5     | 6              | 7      | 8            | 9       | 10 |
|------------|----|-------|-------|-------|----------------|-------|----------------|--------|--------------|---------|----|
|            | A. | A,C,D | A,C,D | A,B,D | B,C            | B,C,D | A,C            | A,C    | A,C          | A,B,C,D | A  |
| SECTION-II | Q. | 1     | 2     | 3     | 4              | 5     | 6              | 7      | 8            |         |    |
|            | A. | 0.60  | 1.33  | 1.50  | 10.10 to 10.16 | 17.66 | 13.95 to 13.96 | 406.00 | 0.40 to 0.42 |         |    |

**PART-2 : CHEMISTRY**

| SECTION-I  | Q. | 1     | 2       | 3     | 4    | 5      | 6     | 7     | 8     | 9   | 10      |
|------------|----|-------|---------|-------|------|--------|-------|-------|-------|-----|---------|
|            | A. | A,B,C | A,B,C,D | A,C,D | C,D  | A      | A,B,C | A,B,C | A,B,C | A,B | A,B,C,D |
| SECTION-II | Q. | 1     | 2       | 3     | 4    | 5      | 6     | 7     | 8     |     |         |
|            | A. | 1.00  | 9.00    | 8.00  | 2.00 | 400.00 | 5.00  | 2.00  | 4.00  |     |         |

**PART-3 : MATHEMATICS**

| SECTION-I  | Q. | 1     | 2    | 3    | 4     | 5    | 6    | 7    | 8    | 9   | 10  |
|------------|----|-------|------|------|-------|------|------|------|------|-----|-----|
|            | A. | A,C,D | B,C  | A,D  | A,C,D | B,C  | B,D  | A,D  | D    | A,C | A,C |
| SECTION-II | Q. | 1     | 2    | 3    | 4     | 5    | 6    | 7    | 8    |     |     |
|            | A. | 4.00  | 0.00 | 2.00 | 6.00  | 0.00 | 8.00 | 7.00 | 3.57 |     |     |

**HINT – SHEET**
**PART-1 : PHYSICS**
**SECTION-I**

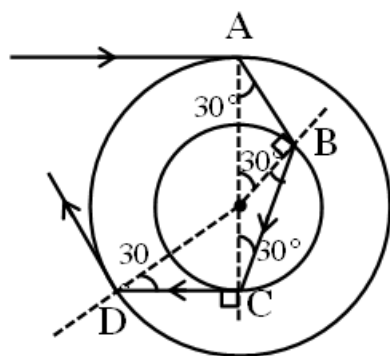
 1. **Ans ( A,C,D )**

$$\phi_1 = \phi_2 = \frac{q}{2\epsilon_0}$$

$$E_1(2\pi R_1^2) = E_2(2\pi R_2^2)$$

 2. **Ans ( A,C,D )**

Apply Snells law at point A, B, C & D. Maximum angle of refraction inside inner sphere is equal to critical angle.


 3. **Ans ( A,B,D )**

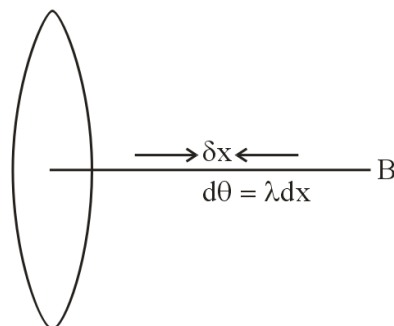
$$\frac{1}{v} - \frac{1}{-v} = \frac{1}{f}$$

$$1 + \frac{v}{u} = \frac{v}{f}$$

$$m = \frac{v}{u} = \frac{v}{f} - 1$$

$$\text{If } m = 0, v = f$$

$$\text{If } v = 0, m = -1$$

 10. **Ans ( A )**


$$\int dU = \int_0^{\sqrt{3}R} \frac{KQ}{\sqrt{R^2 + x^2}} \lambda dx$$

$$U = \frac{Q\lambda}{4\lambda\epsilon_0} \ln(2 + \sqrt{3})$$

## PART-1 : PHYSICS

### SECTION-II

1. **Ans ( 0.60 )**

Apply KVL

In position A

$$10 - i_1 R - 4 = 0$$

$$i_1 = 1.2 \text{ A}$$

In position B

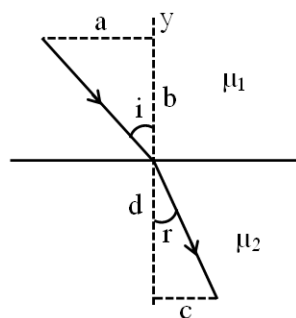
$$10 - i_2 R = 0$$

$$i_2 = 2 \text{ A}$$

$$\therefore \frac{i_1}{i_2} = 0.6 \text{ A}$$

2. **Ans ( 1.33 )**

Snell's law



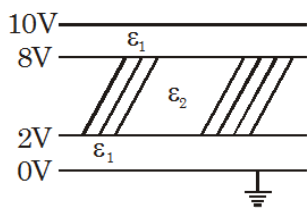
$$\mu_1 \sin i = \mu_2 \sin r$$

$$1.5 \frac{a}{\sqrt{a^2 + b^2}} = 2 \frac{c}{\sqrt{c^2 + d^2}}$$

$$\Rightarrow 1.5 a = 2c$$

$$\frac{a}{c} = \frac{4}{3}$$

3. **Ans ( 1.50 )**



$$Q = CV \left| \begin{array}{l} C_1 = \frac{V_2}{V_1} \\ C_2 = \frac{V_1}{V_2} \end{array} \right.$$

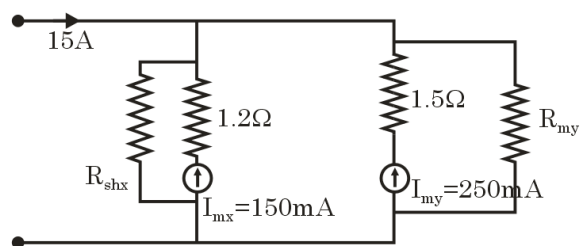
$$C = \frac{A\epsilon}{d} \left| \begin{array}{l} \epsilon_1 = \frac{V_2}{V_1} \\ \epsilon_2 = \frac{V_1}{V_2} \end{array} \right.$$

$$\frac{\epsilon_1}{\epsilon_2} = \frac{V_2}{V_1} \Rightarrow V_2 = \frac{\epsilon_1}{\epsilon_1 + \epsilon_2} (V) \Rightarrow 6 = \frac{\epsilon_1}{\epsilon_1 + \epsilon_2} (10)$$

$$\Rightarrow 3\epsilon_1 + 3\epsilon_2 = 5\epsilon_1 \Rightarrow \frac{\epsilon_1}{\epsilon_2} = \frac{3}{2}$$

$$\epsilon_1 : \epsilon_2 = 3 : 2$$

4. **Ans ( 10.10 to 10.16 )**



X and Y ammeters are connected in parallel

Shunt Registration of X and Y meters :

$$R_{shx} = \frac{1.2}{\left( \frac{15 \times 10^3}{150} - 1 \right)}$$

$$R_{shx} = 0.01212 \Omega$$

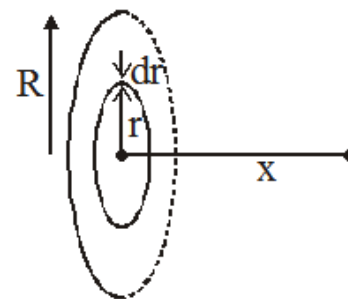
$$R_{shy} = \frac{1.5}{\left( \frac{15 \times 10^3}{250} - 1 \right)}$$

$$R_{shy} = 0.02542 \Omega$$

Current through X ammeter is

$$= \frac{0.02542}{(0.01212 + 0.02542)} \times 15 = 10.157 \text{ amperes}$$

5. **Ans ( 17.66 )**



$$dB = \frac{\mu_0 I r^2}{2(r^2 + r^2)^{\frac{3}{2}}}$$

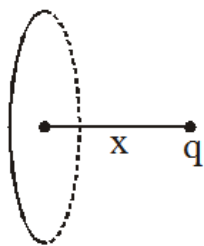
$$I = \sigma r d\omega$$

$$\therefore \int dB = \int_0^R \frac{\mu_0 \sigma r d\omega r^2}{2(r^2 + r^2)^{\frac{3}{2}}}$$

$$= \frac{\mu_0 \sigma^2 \omega}{2} \left[ \frac{1}{r} - \frac{1}{\sqrt{R^2 + r^2}} \right]$$

$$= 176.625 \times 10^{-13} \text{ T}$$

6. Ans ( 13.95 to 13.96 )



$$x = x_1 + x_2$$

COM will not move

$$\therefore mx_1 = Mx_2$$

$$\Rightarrow x_1 \frac{m}{M} = x_2$$

$$F_q = \frac{-kQq(x_1 + x_2)}{[R^2 + (x_1 + x_2)^2]^{\frac{3}{2}}} \quad [\text{For small oscillation,}$$

$$x \ll R]$$

$$\therefore F_q = \frac{-kQq(x_1 + \frac{m}{M}x_1)}{R^3} = \frac{-kQq(\frac{M+m}{M})}{R^3} x_1$$

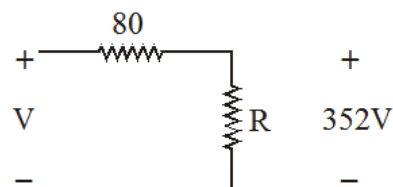
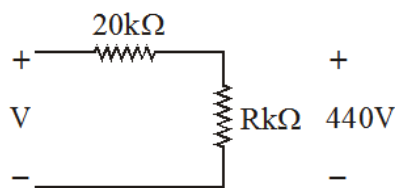
$$\therefore a = -\frac{kQq(M+m)}{mMR^3} x_1$$

$$\therefore T = 2\pi \sqrt{\frac{mMR^3}{kQq(M+m)}}$$

$$= 2\pi \sqrt{\frac{1 \times 2 \times 8}{9 \times 10^9 \times 120 \times 10^{-12} \times 3}} = 13.955 \text{ sec}$$

7. Ans ( 406.00 )

Let resistance of voltmeter be R kΩ



$$V = \left( \frac{440}{R} \right) 20 + 440 \dots (1);$$

$$V = 352 + \left( \frac{352}{R} \right) 80 \dots (2)$$

Solving,  $V = 480$ ;  $R = 220$ ;

$$V_L = \frac{480}{40 + 220} \times 220 = 406V$$

## PART-2 : CHEMISTRY

### SECTION-I

3. Ans ( A,C,D )

For C(s) to be in equilibrium with C(vapours), minimum moles of C(s) formed should be.

$$\frac{0.1 \times 8.2}{0.082 \times 400} = 0.025$$

If moles of C(s) is less than 0.025, C(s) will vaporise completely.

|                      | A(g)       | → | 2B(g) | + | C(s)                           |
|----------------------|------------|---|-------|---|--------------------------------|
| t = 0                | 0.2 mol    |   | 0     |   | 0                              |
| t = t <sub>10%</sub> | 0.2 - 0.02 |   | 0.04  |   | 0.02 < 0.025 hence, C is C(g)  |
| t = t <sub>5%</sub>  | 0.2 - 0.01 |   | 0.02  |   | 0.01 < 0.025, hence, C is C(g) |
| t = 20 min           | 0.2 - 0.1  |   | 0.2   |   | 0.1 > 0.025                    |
| t = ∞                | 0          |   | 0.4   |   | 0.4 > 0.025 ,                  |

pressure due to C = 0.1 atm

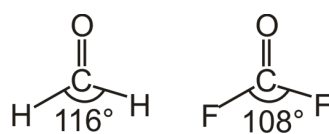
$$\text{At } t_{10\%}, P = \frac{0.24 \times 0.082 \times 400}{8.2} = 0.96 \text{ atm}$$

$$t_{5\%}, P = \frac{0.22 \times 0.082 \times 400}{8.2} = 0.88 \text{ atm}$$

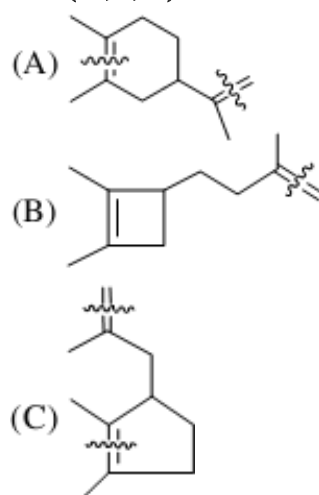
$$t_{\infty}, P = \frac{0.4 \times 0.082 \times 400}{8.2} + 0.1 = 1.7 \text{ atm}$$

$$t_{20\text{min}} P = \frac{0.3 \times 0.082 \times 400}{8.2} + 0.1 = 1.3 \text{ atm}$$

4. Ans ( C,D )



8. Ans ( A,B,C )



## PART-2 : CHEMISTRY

### SECTION-II

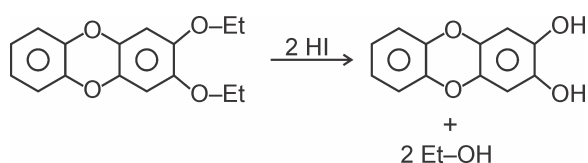
3. **Ans ( 8.00 )**

$$\Delta S_{\text{water}} = \frac{3000}{300} = 10 \text{ J/K}$$

$$\Delta S_{\text{fish}} = \frac{-2}{3} \times 10^{-3} \times 3000 = -2 \text{ J/K}$$

$$\Delta S_{\text{total}} = 8 \text{ J/K}$$

7. **Ans ( 2.00 )**



## PART-3 : MATHEMATICS

### SECTION-I

1. **Ans ( A,C,D )**

$$\text{for A, B } \int x \left(1 - \frac{1}{x^4}\right)^{1/5} \frac{1}{x^6} dx$$

$$\text{let } 1 - \frac{1}{x^4} = t \Rightarrow dt = \frac{4}{x^5} dx$$

$$\int t^{1/5} \frac{dt}{4} = \frac{1}{4} \frac{t^{6/5}}{6/5} = \frac{5}{24} t^{6/5}$$

$$\begin{aligned} \text{(C)} \int (\tan^2 x + 1) (\tan x + 1) dx \\ \frac{(\tan x + 1)^2}{2} + C \end{aligned}$$

$$\begin{aligned} \text{(D)} \int (\tan x \sec^2 x + \sec^2 x) dx \\ \frac{1}{2} \tan^2 x + \tan x + C \\ \text{or } \frac{\sec^2 x}{2} + \tan x + \left(C - \frac{1}{2}\right) \end{aligned}$$

2. **Ans ( B,C )**

$$I_n = \int_0^1 (1 - x^2)^n dx \text{ Put } x = \cos \theta$$

$$\Rightarrow I_n = \int_0^{\pi/2} \sin^{2n+1} \theta d\theta$$

use walli's formula

$$I_n = \frac{2n}{2n+1} \cdot \frac{2n-2}{2n-1} \cdot \frac{2n-4}{2n-3} \cdots \frac{4}{5} \cdot \frac{2}{3} \quad \dots(1)$$

$$I_{n+1} = \frac{2n+2}{2n+3} \cdot \frac{2n}{2n+1} \cdots \frac{4}{5} \cdot \frac{2}{3} \quad \dots(2)$$

$$\frac{I_{n+1}}{I_n} = \frac{2n+2}{2n+3} \text{ and } (1) \Rightarrow I_k = \sum_{r=1}^k \frac{2r}{2r+1}$$

Also in  $I_n$ , each fraction is less than 1 so  $I_n < 1 \forall n \in \mathbb{N}$

3. **Ans ( A,D )**

$$\int \sqrt{1 + \cos x} dx = \int \frac{\cos x}{\sqrt{\sin x - \sin^2 x}} dx \quad \dots(1)$$

$$\text{Let } \sin x = t, \int \frac{dt}{\sqrt{t-t^2}} = \int \frac{dt}{\sqrt{\frac{1}{4} - (t-\frac{1}{2})^2}}$$

$$= \sin^{-1}(2t-1) + C$$

$$= C - \sin^{-1}(1-2\sin x)$$

$$= C + \cos^{-1}(1-2\sin x)$$

OR

$$\text{Let } \sin x = t^2$$

from (1)

$$\begin{aligned} \int \frac{2t dt}{\sqrt{t^2 - t^4}} &= 2 \int \frac{dt}{\sqrt{1-t^2}} \\ &= 2\sin^{-1} t + C = 2\sin^{-1}(\sqrt{\sin x}) + C \end{aligned}$$

4. **Ans ( A,C,D )**

$$A_n = \sum_{t=1}^n \frac{n}{n^2 + nt + t^2} \text{ put } k+1 = t$$

$$B_n = \sum_{k=0}^{n-1} \frac{n}{n^2 + kx + k^2},$$

$$\text{now on solve } A_n < \frac{\pi}{3\sqrt{3}} < B_n$$

5. Ans (B,C)

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{\substack{x \rightarrow 1^+ \\ n \rightarrow \infty}} \frac{2 + \frac{x}{x^{2n}} + \frac{2}{x^{2n}}}{1 - \frac{x^2}{x^{2n}} + \frac{3}{x^{2n}}} = 2$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{\substack{x \rightarrow 1^- \\ n \rightarrow \infty}} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 2} = \lim_{x \rightarrow 1^-} \frac{x + 2}{-x^2 + 2} = \frac{3}{1} = 3$$

$\therefore \text{LHL} \neq \text{RHL}$

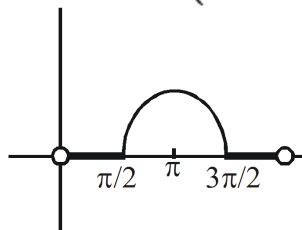
$\therefore$  Discontinuous at  $x = 1$ .

Similarly

$$\lim_{x \rightarrow 2} f(x) = 2; \lim_{x \rightarrow -2} f(x) = 2$$

6. Ans (B,D)

$$\cos x [\cos x] = \begin{cases} 0 & 0 < x \leq \frac{\pi}{2} \\ -\cos x & \frac{\pi}{2} < x \leq \pi \\ -\cos x & \pi < x < \frac{3\pi}{2} \\ 0 & \frac{3\pi}{2} \leq x < 2\pi \end{cases}$$



7. Ans (A,D)

$P(\alpha, \beta)$

coc is  $\beta y = 2(x + \alpha)$

$$y = \frac{2}{\beta}x + \frac{2\alpha}{\beta} \text{ is normal}$$

$$\frac{2\alpha}{\beta} = -2 \left( \frac{2}{\beta} \right) - \left( \frac{2}{\beta} \right)^3$$

$$\Rightarrow \alpha\beta^2 = -2\beta^2 - 4$$

$$\Rightarrow (\alpha + 2)\beta^2 + 4 = 0$$

$$\text{Locus : } (x + 2)y^2 + 4 = 0$$

$$(\alpha, \beta) \in I \quad \alpha = -2 - \frac{4}{\beta^2}$$

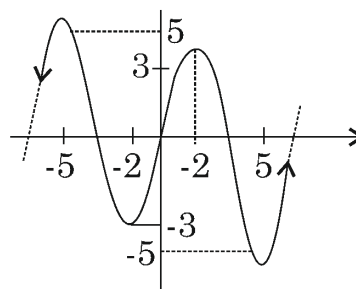
$$\therefore \beta^2 = 1, 4$$

$$\beta = 1, -1, 2, -2$$

$$\alpha = -6, -6, -3, -3$$

total 4 pairs

8. Ans (D)



$$f(x) = ax^9 + b \sin x + cx^2 \operatorname{sgn}(x) + \frac{e^x - e^{-x}}{e^x + e^{-x}};$$

$a > 0$  &  $b, c \in \mathbb{R}$

is an odd function (check yourself)

$$\Rightarrow f(0) = 0$$

$$f(-5) = 5, f(-2) = -3 \Rightarrow f(5) = -5, f(2) = 3$$

Also, as  $x \rightarrow \infty$ ;  $f(x) \rightarrow \infty$  and  $x \rightarrow -\infty$ ;  $f(x) \rightarrow -\infty$

Hence no. of zeroes of  $f(x) = 0$  is 5

9. Ans (A,C)

Differentiating  $fg = 1$ , we get  $f'g + g'f = 0$

$$\Rightarrow \frac{f'}{f} = -\frac{g'}{g}$$

Differentiating again, we get  $f''g + 2f'g' + g''f = 0$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 + \frac{g''}{g'} \cdot \frac{f}{f'} = 0$$

$$\Rightarrow \frac{f''}{f'} \cdot \frac{g}{g'} + 2 - \frac{g''}{g'} \cdot \frac{g}{g'} = 0 \Rightarrow \frac{g}{g'} \left( \frac{f''}{f'} - \frac{g''}{g'} \right) = -2$$

$$\Rightarrow \left( \frac{f''}{f'} - \frac{g''}{g'} \right) = \frac{-2g'}{g} \text{ or } \frac{2f'}{f}$$

10. Ans (A,C)

$$\int \frac{7 + 6x^{13}}{\sqrt[3]{x^{17} - x^{30}}} dx = \int \frac{7x^{-8} + 6x^5}{\sqrt[3]{x^{-7} - x^6}} dx$$

$$\text{put } x^{-7} - x^6 = t^3$$

$$(-7x^{-8} - 6x^5) dx = 3t^2 dt$$

$$(7x^{-8} + 6x^5) dx = -3t^2 dt$$

$$\therefore I = \int \frac{-3t^2}{t} dt = - \int 3t dt = \frac{-3}{2} t^2 + C$$

$$= \frac{-3}{2} (x^{-7} - x^6)^{\frac{2}{3}} = -\frac{3}{2} \frac{(1 - x^{13})^{\frac{2}{3}}}{x^{\frac{14}{3}}} + C$$

## PART-3 : MATHEMATICS

### SECTION-II

1. **Ans ( 4.00 )**

$$A = \int_0^1 \frac{e^x dx}{\sqrt{1-x^2}}$$

Put  $x = \sin \theta$

$$\Rightarrow A = \int_0^{\pi/2} e^{\sin \theta} d\theta$$

$$\text{Let } I = \int_0^{\pi} (e^{|\sin x|} + e^{|\cos x|}) dx$$

↓

$$\text{Period} = \frac{\pi}{2} \Rightarrow I = 2 \int_0^{\pi/2} (e^{|\sin x|} + e^{|\cos x|}) dx$$

$$\Rightarrow I = 4 \int_0^{\pi/2} e^{|\sin x|} dx = 4A$$

2. **Ans ( 0.00 )**

$$\text{Using } \int_0^1 f(x) dx = \int_0^1 f(1-x) dx$$

$$I = \int_0^1 \sqrt[3]{x^2(2x-3) + (1-x)} dx$$

$$= \int_0^1 \sqrt[3]{(1-x)^2(-1-2x) + x} dx$$

$$= - \int_0^1 \sqrt[3]{(x^2-2x+1)(1+2x) - x} dx$$

$$= - \int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1} dx = -I$$

$$2I = 0 \quad \therefore I = 0$$

3. **Ans ( 2.00 )**

$$\int \frac{(3x^2 + 2x)dx}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5}$$

$$= \frac{1}{2} \tan^{-1} \left( \frac{x^3 + x^2 + 1}{2} \right) + C$$

$$\therefore a = 2 \text{ and } g(1) = \frac{3}{2}, \quad g(0) = \frac{1}{2}$$

4. **Ans ( 6.00 )**

$$f''(0) = 2; \lim_{x \rightarrow 0} \frac{f(5x) - 3f(3x) + 2f(2x)}{x^2}$$

Applying L.hospital twice

$$\lim_{x \rightarrow 0} \frac{25f''(5x) - 27f''(3x) + 8f''(2x)}{2}$$

$$= \frac{6f''(0)}{2} = 6$$

5. **Ans ( 0.00 )**

$$g(x) > 0 \Rightarrow f(x) > 0$$

$$g\left(\frac{1}{2}\right) = e \Rightarrow f\left(\frac{1}{2}\right) = e$$

$$\frac{dg(x)}{dx} < 0 \Rightarrow f'(x) < 2f(x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} < 2$$

$$\Rightarrow \int_{1/2}^x \frac{f'(x)}{f(x)} dx < \int_{1/2}^x 2 dx$$

$$\therefore 0 < f(x) < e^{2x-1}$$

$$\int_{1/2}^1 0 dx < \int_{1/2}^1 f(x) dx < \int_{1/2}^1 e^{2x-1} dx$$

$$0 < \int_{1/2}^1 f(x) dx < \frac{e-1}{2}$$

6. **Ans ( 8.00 )**

$f(x)$  is discontinuous at

$$x = \pm\sqrt{3}, \pm\sqrt{2}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$$

only since  $\sin \pi x = 0$  at  $x = \pm 1, \pm 2, 0, 3$  are points where  $f(x)$  is continuous.

7. Ans ( 7.00 )

At  $A = \frac{\pi}{7}$ ,  $B = \frac{2\pi}{7}$ ,  $C = \frac{4\pi}{7}$  which are angles

of a triangle

So,  $x = \tan A$ ,  $y = \tan B$ ,  $z = -\tan C$

$$\begin{aligned}
 & \frac{1}{2}(x^2 + y^2 + z^2 - x^2y^2z^2) \\
 &= \frac{1}{2} [\tan^2 A + \tan^2 B + \tan^2 C - (\tan A \tan B \tan C)^2] \\
 &= -(\tan A \tan B + \tan B \tan C + \tan C \tan A) \\
 &= -\frac{\sin A \sin B \cos C + \sin B \sin C \cos A + \sin C \sin A \cos B}{\cos A \cos B \cos C} \\
 &= \frac{\cos(A + B + C) - \cos A \cos B \cos C}{\cos A \cos B \cos C} \\
 &= -\frac{1}{\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}} - 1 = 8 - 1 = 7
 \end{aligned}$$

8. Ans ( 3.57 )

$$\therefore y = \frac{2x^2 + 7x + 3}{2x^2 + x + 1} \quad \dots(i)$$

$$\Rightarrow 2x^2y + xy + y = 2x^2 + 7x + 3$$

$$\Rightarrow 2x^2(y - 1) + x(y - 7) + (y - 3) = 0$$

$$\text{if } y - 1 \neq 0$$

$$\therefore x \in \mathbb{R} \Rightarrow D \geq 0$$

$$(y - 7)^2 - 8(y - 1)(y - 3) \geq 0$$

$$\Rightarrow y^2 - 14y + 49 - 8y^2 + 32y - 24 \geq 0$$

$$\Rightarrow -7y^2 + 18y + 25 \geq 0$$

$$\Rightarrow 7y^2 - 18y - 25 \leq 0$$

$$\Rightarrow (y + 1)(7y - 25) \leq 0$$

$$-1 \leq y \leq \frac{25}{7} \quad \text{but } y \neq 1$$

$$\text{put } y = 1 \text{ in (1)} \quad \therefore (1)$$

$$\Rightarrow 2x^2 + 7x + 3 = 2x^2 + x + 1$$

$$\Rightarrow x = -\frac{1}{3}$$



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

10-07-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

ANSWER KEY

PAPER-2

### PART-1 : PHYSICS

| SECTION-I (i)  | Q. | 1     | 2    | 3     | 4     | 5      | 6     |
|----------------|----|-------|------|-------|-------|--------|-------|
|                | A. | A     | B    | A     | D     | C      | D     |
| SECTION-I (ii) | Q. | 7     | 8    | 9     | 10    | 11     | 12    |
|                | A. | B,C   | C    | A     | A     | C      | D     |
| SECTION-II     | Q. | 1     | 2    | 3     | 4     | 5      | 6     |
|                | A. | 81.00 | 9.00 | 26.00 | 21.00 | 300.00 | 30.76 |

### PART-2 : CHEMISTRY

| SECTION-I (i)  | Q. | 1     | 2    | 3     | 4     | 5     | 6     |
|----------------|----|-------|------|-------|-------|-------|-------|
|                | A. | B     | A    | D     | C     | B     | D     |
| SECTION-I (ii) | Q. | 7     | 8    | 9     | 10    | 11    | 12    |
|                | A. | D     | A    | A,C,D | B,D   | A,B,C | A,B,C |
| SECTION-II     | Q. | 1     | 2    | 3     | 4     | 5     | 6     |
|                | A. | 15.00 | 9.00 | 4.00  | 64.00 | 0.60  | 2.00  |

### PART-3 : MATHEMATICS

| SECTION-I (i)  | Q. | 1     | 2       | 3       | 4            | 5    | 6    |
|----------------|----|-------|---------|---------|--------------|------|------|
|                | A. | C     | C       | A       | B            | A    | D    |
| SECTION-I (ii) | Q. | 7     | 8       | 9       | 10           | 11   | 12   |
|                | A. | A,C,D | A,B,C,D | A,B,C,D | A            | A    | A,D  |
| SECTION-II     | Q. | 1     | 2       | 3       | 4            | 5    | 6    |
|                | A. | 1.00  | 5.00    | 6.00    | 0.41 to 0.42 | 2.00 | 2.11 |

### HINT – SHEET

#### PART-1 : PHYSICS

##### SECTION-I (i)

1. Ans (A)

Equation of straight line according to graph:

$$P = - \left( \frac{P_0}{V_0} \right) V + 2P_0$$

$$PV = nRT$$

$$\left( -\frac{P_0}{V_0} V + 2P_0 \right) V = nRT$$

$$T = -\frac{P_0}{V_0 R} V^2 + \frac{2P_0}{R} V \dots (i)$$

$$\frac{dT}{dV} = 0 \Rightarrow V = V_0$$

$$\Rightarrow T_0 = \frac{P_0 V_0}{nR} (T_{\max})$$

$$= \left( \frac{P_0 V_0}{R} \right)$$

$$\text{At B, } T \rightarrow \frac{T_0}{2}$$

$$\frac{T_0}{2} = \frac{P_0 V_0}{2R} \dots (ii)$$

By (i) &amp; (ii)

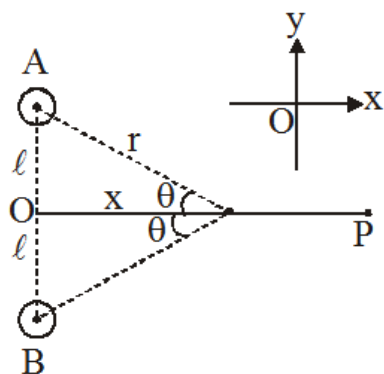
$$\frac{T_0}{2} = \frac{P_0 V_0}{2R} = -\frac{P_0}{V_0 R} V^2 + \frac{2P_0}{R} V$$

$$\frac{T_0}{2} = -\frac{P^2}{V_0} + 2V$$

$$2V^2 - 4V_0 V + V_0^2 = 0$$

$$V = \left( \frac{\sqrt{2} + 1}{\sqrt{2}} \right) V_0$$

2. Ans (B)



Net B at distance x from O along x-axis

$$= \frac{\mu_0 I}{2\pi r} 2 \cos \theta$$

$$\Rightarrow F = \int dF = \int l \frac{\mu_0 I}{2\pi r} 2 \cos \theta dx$$

$$= \frac{\mu_0}{2\pi} I^2 \ell \ln \left( 1 + \frac{L^2}{\ell^2} \right)$$

3. Ans (A)

Using conservation of angular momentum about fixed charged.

$$MV \sin 30^\circ R = Mv' r \Rightarrow \frac{V}{2} R = \frac{V}{\sqrt{3}} r$$

$$r = \frac{\sqrt{3}}{2} R$$

4. Ans (D)

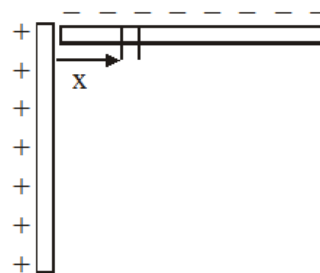
$$\text{Using } \frac{1}{f} = (\mu_r - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$f > 0 \text{ for lower half of lens as } \mu_r = \frac{\mu_1}{\mu_3} > 1$$

$$f < 0 \text{ for upper half of lens as } \mu_r = \frac{\mu_1}{\mu_2} < 1$$

So, lower half acts as convergent lens and upper half as divergent lens.

5. Ans (C)



At line AB, potential =  $V_0$

$$U_i = \lambda L V_0$$

$$U_f = \int_0^L V_x \lambda dx$$

$$V(x) = V_0 - E_0 x$$

$$U_f = \lambda V_0 L - \lambda E_0 \frac{L^2}{2}$$

$$U_i - U_f = \lambda E_0 \frac{L^2}{2} \quad \dots(i)$$

In equilibrium:

$$F_e = \lambda L E_0$$

$$\tau_{Fe} = \tau_{mg} \text{ (Torque about A)}$$

$$\lambda E_0 \frac{L}{2} \cos \theta = mg \frac{L}{2} \sin \theta \quad \dots(ii)$$

By equation (i) & (ii)

$$\frac{\lambda L^2 E_0}{2} \cos \theta = \frac{\lambda L^2 E_0}{2} \sin \theta$$

$$\tan \theta = 1, \theta = 45^\circ$$

Applying energy conservation

KE = loss in electrostatic PE – gain in gravitational PE

$$= \int_0^L E_0 x \sin \theta \lambda dx - mg \frac{L}{2} (1 - \cos \theta)$$

$$\frac{E_0 x L^2}{2\sqrt{2}} - mg \frac{1}{2} \left( 1 - \frac{1}{\sqrt{2}} \right)$$

$$\frac{mgL}{2\sqrt{2}} - \frac{mgL}{2} \left( 1 - \frac{1}{\sqrt{2}} \right) = \frac{1}{2} \frac{ML^2}{3} \omega^2$$

$$\text{Solving } \omega = \sqrt{\frac{(3\sqrt{2} - 1)g}{L}}$$

6. **Ans (D)**

KVL : At  $t = 0$ ,

$$iR + \frac{q_0}{C_0} - E = 0 \Rightarrow i = \frac{E - q_0/C_0}{R}$$

At  $t \rightarrow \infty$ ,  $i = 0$

$$\therefore \frac{q_0 + q}{C_0} + \frac{q}{C} - E = 0$$

$$\Rightarrow q_C = q = \frac{C_0 C E}{C_0 + C} \left( 1 - \frac{q_0}{C_0 E} \right)$$

**PART-1 : PHYSICS**

**SECTION-I (ii)**

7. **Ans (B,C)**

Applying Bernoulli's theorem between top and bottom of chimney

$$P + \rho_s g h + \frac{1}{2} \rho_s v^2 = P_0, \text{ where } v = Q/A$$

$$\text{So, } P = P_0 - \rho_s g h - \frac{1}{2} \rho_s \frac{Q^2}{A^2} = P_0 - \rho_a g h$$

$$\text{So, } h = \frac{\rho_s Q^2}{2(\rho_a - \rho_s) A^2 g}$$

$$\text{For ideal gas, } \frac{\rho_a}{\rho_s} = \frac{T_s}{T_a}, \text{ So } h = \frac{T_a Q^2}{2g A^2 \Delta T}$$

$$\frac{h_w}{h_c} = \frac{(273 + 27)(327 + 23)}{(327 - 27)(273 - 23)}, \text{ where } h_c = 100 \text{ m.}$$

$$\text{So, } h_w = 140 \text{ m}$$

8. **Ans (C)**

Applying Bernoulli's theorem between top and bottom of chimney

$$P + \rho_s g h + \frac{1}{2} \rho_s v^2 = P_0, \text{ where } v = Q/A$$

$$\text{So, } P = P_0 - \rho_s g h - \frac{1}{2} \rho_s \frac{Q^2}{A^2} = P_0 - \rho_a g h$$

$$\text{So, } h = \frac{\rho_s Q^2}{2(\rho_a - \rho_s) A^2 g}$$

$$\text{For ideal gas, } \frac{\rho_a}{\rho_s} = \frac{T_s}{T_a}, \text{ So } h = \frac{T_a Q^2}{2g A^2 \Delta T}$$

$$\frac{h_w}{h_c} = \frac{(273 + 27)(327 + 23)}{(327 - 27)(273 - 23)}, \text{ where } h_c = 100 \text{ m.}$$

$$\text{So, } h_w = 140 \text{ m}$$

9. **Ans (A)**

$$m = \frac{f}{f - u} = \frac{h_1}{h_0}$$

$$\Rightarrow h_1 = \frac{-15}{-15 + 50} \times 1 = \frac{-3}{7} \text{ mm}$$

$$\therefore \text{Amplitude of image} = |h_1| = \frac{3}{7} \text{ mm.}$$

10. **Ans (A)**

Velocity of image and object are in opposite directions.

11. **Ans (C)**

$$\Delta Q_{AB} = n C_p \Delta T = 2 \times \frac{5R}{2} (900 - 300) = 3000R$$

In the process 'BC',

VT = constant

PV = constant

$$\therefore C = C_v + \frac{R}{1 - x} = \frac{3R}{2} + \frac{R}{1 - 2} = \frac{R}{2}$$

$$\therefore \Delta Q_{BC} = n C \Delta T = \frac{2R}{2} (300 - 900) = -600R$$

$$\Delta W_{BC} = \frac{n R \Delta T}{(1 - x)} - \frac{2R(300 - 900)}{(1 - 2)} = 1200R$$

$$\Delta Q_{CA} = n R T_0 \ln \left( \frac{V_0}{9V_0} \right) = 2R \times 300 \ln \left( \frac{1}{9} \right)$$

$$= -1200 \ln(3) = -1200 R \times 1.1 = -1320 R$$

$$\therefore \Delta W_{\text{cycle}} = \Delta Q_{\text{cycle}} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA}$$

$$= 3000 R - 600 R - 1320 R = 1080 R$$

$\therefore$  Efficiency of cycle,

$$\eta = \frac{\Delta W_{\text{cycle}}}{\Delta Q_{\text{supplied}}} \times 100 = \frac{1080R}{3000R} \times 100 = 36\%$$

12. **Ans (D)**

$$\Delta Q_{AB} = n C_p \Delta T = 2 \times$$

$$\frac{5R}{2} (900 - 300) = 3000R$$

In the process 'BC',

VT = constant

PV = constant

$$\therefore C = C_v + \frac{R}{1 - x} = \frac{3R}{2} + \frac{R}{1 - 2} = \frac{R}{2}$$

$$\therefore \Delta Q_{BC} = n C \Delta T = \frac{2R}{2} (300 - 900) = -600R$$

$$\Delta W_{BC} = \frac{n R \Delta T}{(1 - x)} - \frac{2R(300 - 900)}{(1 - 2)} = 1200R$$

$$\Delta Q_{CA} = n R T_0 \ln \left( \frac{V_0}{9V_0} \right) = 2R \times 300 \ln \left( \frac{1}{9} \right)$$

$$= -1200 \ln(3) = -1200 R \times 1.1 = -1320 R$$

$$\therefore \Delta W_{\text{cycle}} = \Delta Q_{\text{cycle}} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA}$$

$$= 3000 R - 600 R - 1320 R = 1080 R$$

$\therefore$  Efficiency of cycle,

$$\eta = \frac{\Delta W_{\text{cycle}}}{\Delta Q_{\text{supplied}}} \times 100 = \frac{1080R}{3000R} \times 100 = 36\%$$

## PART-1 : PHYSICS

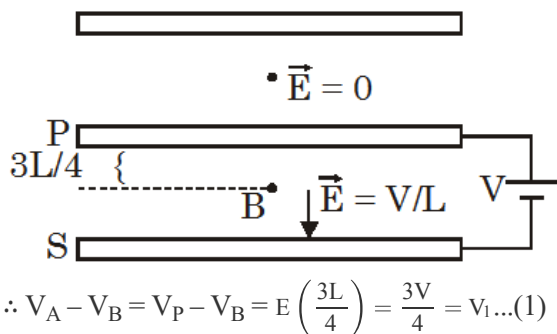
### SECTION-II

1. **Ans ( 81.00 )**

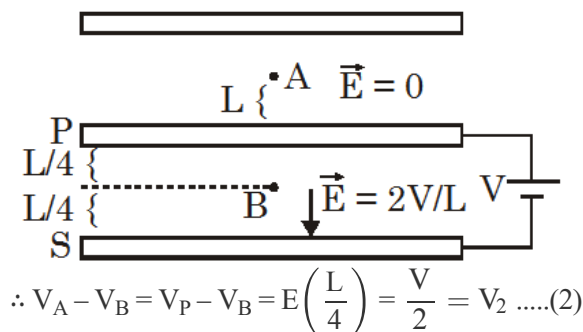
$$\text{Heat} = \varepsilon \times \left( \frac{3C\varepsilon}{2} \right) - \left( \frac{c^2\varepsilon^2}{4C} + \frac{c^2\varepsilon^2}{8C} - 0 \right)$$

2. **Ans ( 9.00 )**

Initially



Later,



from (1) and (2)

$$\frac{V_1}{V_2} = \frac{6}{4} \Rightarrow \frac{6V_1}{V_2} = 9$$

3. **Ans ( 26.00 )**

By convex lens  $v = +20\text{cm}$

By concave lens  $u = -10\text{ cm}$

$$\frac{1}{v} - \frac{1}{-10} = \frac{1}{-10}$$

$v = -5\text{cm}$

So, x-coordinate is  $(30 - 5)\text{ cm} = 25\text{ cm}$  ;

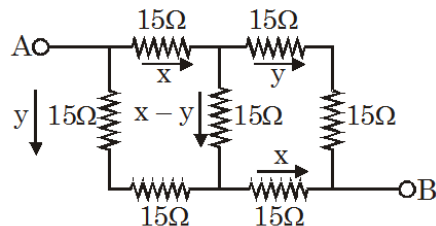
$$m = \frac{v}{u} = \frac{h_1}{5\text{mm}} \Rightarrow h_1 = 2.5\text{ mm} = 0.25\text{ cm}$$

So, y-coordinate is  $+0.25\text{ cm}$

4. **Ans ( 21.00 )**

Let  $V_{AB} = V \Rightarrow$  by KVL,

$$15x + 30y = V \dots\dots (1)$$



and  $15x + 15(x - y) + 15x = V \dots\dots (ii)$

Solving (i) & (ii)

$$x + y = \frac{5V}{105} \Rightarrow R = \frac{V}{x + y} = 21\Omega$$

5. **Ans ( 300.00 )**

Assume cavity of  $\rho$  density

$$\text{So } V_c = \frac{3}{2} \frac{k\rho \times \frac{4}{3}\pi R^3}{R} + \frac{k(-\rho) \times \frac{4}{3}\pi \left(\frac{R}{2}\right)^3}{R/2}$$

6. **Ans ( 30.76 )**

From energy conservation

$$\frac{-G \times 1 \times 2}{1} = \frac{-G \times 1 \times 2}{1} + \frac{1}{2} \times 1 \times (2 \times 10^{-5})^2 + \frac{1}{2} \times 2 \times (1 \times 10^{-5})^2 \dots (i)$$

From momentum conservation

$$2v_B = 1v_A$$

$$2 \times \frac{3.6 \times 10^{-3}}{100} = v_A$$

$$v_A = 2 \times 10^{-5} \text{ m/s}$$

$$v_B = 10^{-5} \text{ m/s}$$

$$r = 30.76 \text{ cm}$$

## PART-2 : CHEMISTRY

### SECTION-I (i)

3. **Ans ( D )**

$$Z = \frac{PV_m}{RT} \Rightarrow 0.5 = \frac{24 \times V_m}{24}$$

$$V_m = 0.5 \text{ l/mole}$$

$$\left( P + \frac{a}{V_m^2} \right) (V_m - b) = RT$$

$$\left( 24 + \frac{a}{0.5^2} \right) (0.5 - 0.1) = 24$$

$$24 + \frac{a}{0.5^2} = 60$$

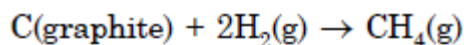
$$\frac{a}{0.5^2} = 36$$

$$a = 36 \times 0.5^2 = 9 \frac{\text{atm} \cdot \ell^2}{\text{mol}^2}$$

## PART-2 : CHEMISTRY

### SECTION-I (ii)

11. Ans (A,B,C)



$$\Delta_f H^\circ = \{-400 + 2(-300)\} - \{-900\}$$

$$\Delta_f H^\circ = -100$$

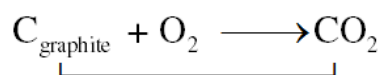
$$\Delta_f H^\circ = -100 = 700 + 880 - 4E_{\text{C-H}}$$

$$E_{\text{(C-H)}} = 420 \frac{\text{kJ}}{\text{mole}}$$

12. Ans (A,B,C)

As per Question:

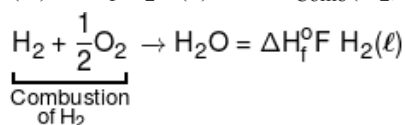
$$(A) \Delta H^\circ_{\text{Cumb}}(\text{C}_{\text{graphite}}) = \Delta H^\circ_f(\text{CO}_2, \text{g})$$



From Reaction we can say that

Option A is correct

$$(B) \Delta H^\circ_f \text{H}_2\text{O}(\ell) = \Delta H^\circ_{\text{Comb}}(\text{H}_2, \text{g})$$



Option B is also correct

(C)  $\Delta H^\circ_{\text{Combustion}}$  is always Negative as it releases energy and bond gets Break.

(D)  $\Delta H^\circ_f$  i.e. heat of Formation cannot be always Negative.

## PART-2 : CHEMISTRY

### SECTION-II

1. Ans (15.00)

If M is the molar mass of solute, we will have

$$20 \text{ mmHg} = \frac{(90\text{g}/18\text{gmol}^{-1})}{(90\text{g}/18\text{gmol}^{-1}) + (30\text{g}/M)} p_1^* \dots\dots(1)$$

$$21 \text{ mmHg} = \frac{(108\text{g}/18\text{gmol}^{-1})}{(108\text{g}/18\text{gmol}^{-1}) + (30\text{g}/M)} p_1^* \dots\dots(2)$$

Dividing Eq. (2) by Eq.(1) we get

$$\frac{21.00}{20.00} = \frac{6}{6\text{mol} + (30 + \text{g}/M)} \cdot \frac{5\text{mol} + (30\text{g}/M)}{5\text{mol}}$$

$$= 15.00 \text{ g mol}^{-1}$$

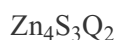
2. Ans (9.00)

$$\text{Zn} = 4$$

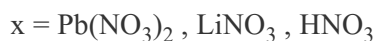
$$\text{S}^{-2} = \frac{1}{8} \times 4 + \frac{1}{2} \times 5 = 3$$

Total positive charge that must be compensated

$$\text{is} = +2.$$

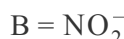


5. Ans (0.60)



$$= \frac{3}{5} = 0.60$$

6. Ans (2.00)



## PART-3 : MATHEMATICS

### SECTION-I (i)

1. Ans (C)

$$(\theta_1 + \theta_2) \left( \frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} \right) = 1$$

$$\cos^4 x + \sin^4 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = 1$$

$$1 - 2\sin^2 x \cos^2 x + \frac{\theta_2}{\theta_1} \cos^4 x + \frac{\theta_1}{\theta_2} \sin^4 x = 1$$

$$\left( \sqrt{\frac{\theta_2}{\theta_1}} \cos^2 x - \sqrt{\frac{\theta_1}{\theta_2}} \sin^2 x \right)^2 = 0$$

$$\tan^2 x = \frac{\theta_2}{\theta_1}$$

$$\theta_2 = \theta_1 \tan^2 x$$

$$\frac{d\theta_2}{d\theta_1} = \tan^2 x$$

2. Ans (C)

Given

$$F(x) = C \sin x \int_0^x \cos t \, dt + 2 \int_0^x t \, dt + C \cos^2 x - x^2$$

$\Rightarrow F'(x) = 0 \therefore F(x)$  is a constant function

$$\therefore F(x) = F(0) = C$$

Given

$$x^2 - 2x + 3 > C \, \forall x \Rightarrow x^2 - 2x + (3 - 1) > 0 \, \forall x$$

$$\Rightarrow D \leq 0 \Rightarrow 4 - 4(3 - C) \leq 0$$

$$\Rightarrow 1 - 3 + C \leq 0 \Rightarrow C \leq 2$$

$$\text{Now, } \left( \frac{6e^x - e^{2x} - 1}{e^x} \right)^{f(2)} = \left( \underbrace{6 - \left( e^x + \frac{1}{e^x} \right)}_{\leq 4} \right)^c \leq 4^2$$

3. Ans (A)

Using LMVT for  $f$  in  $[1, 2]$ , we have for  $c \in (1, 2)$

$$\frac{f(2) - f(1)}{2 - 1} = f'(c) \leq 2$$

$$f(2) - f(1) \leq 2 \Rightarrow f(2) \leq 4 \quad \dots(A) \quad (\because f(1) = 2)$$

Again, using LMVT in  $[2, 4]$ , we have for  $d \in (2, 4)$

$$\frac{f(4) - f(2)}{4 - 2} = f'(d) \leq 2$$

$$f(4) - f(2) \leq 4 \Rightarrow f(2) \geq 4 \quad \dots(B) \quad (\because f(4) = 8)$$

From Equation (A) and (B)

We get  $f(2) = 4$

4. Ans (B)

$$\lim_{t \rightarrow x+1} \frac{2t f(x+1) - (x+1)^2 f'(t)}{f'(t)} = 1$$

$$\Rightarrow \frac{2(x+1)f(x+1) - (x+1)^2 f'(x+1)}{f'(x+1)} = 1$$

$$\Rightarrow 2x f(x) - x^2 f'(x) = f'(x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{2x}{1+x^2}$$

$$\therefore \lim_{x \rightarrow 1} = \frac{2x}{1+x^2}$$

$$\therefore \lim_{x \rightarrow 1} \frac{\ln(f(x)) - \ln 2}{x - 1}$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{\frac{f'(x)}{f(x)}}{1} \Rightarrow 1$$

5. Ans (A)

$$f'\left(\frac{\pi}{3}\right) = 0 + 0 + \left(\sin \frac{\pi}{3}\right)^{1/3} \cdot \left(\sin \frac{2\pi}{3}\right)^{1/2} \cdot \left(\cos \frac{3\pi}{3}\right) \cdot 3$$

$$= \left(\frac{\sqrt{3}}{2}\right)^{1/3} \cdot \left(\frac{\sqrt{3}}{2}\right)^{1/2} \cdot (-3) = \frac{-3^{17/12}}{2^{5/6}}$$

6. Ans (D)

$$I = \int \frac{x^2 + 20}{(x \sin x + 5 \cos x)^2} dx \text{ multiply \& div by } x^8$$

$$I = \int \frac{x^5 \cdot x^3 (x^2 + 20)}{(x^5 \sin x + 5x^4 \cos x)^2} dx$$

$$I = \int \frac{\left(\frac{x^5}{\cos x}\right) x^3 (x^2 + 20) \cos x}{\frac{(x^5 \sin x + 5x^4 \cos x)^2}{I}} dx$$

$$x^5 \sin x + 5x^4 \cos x = t$$

$$[x^5 \cos x + \sin x 5x^4 + 5x^4 (-\sin x) + 5 \cos x \times 4x^3]$$

$$dx = dt$$

$$x^3 \cos x (x^2 + 20) dx = dt$$

$$I = \frac{x^5}{\cos x} \frac{1}{(x^5 \sin x + 5x^4 \cos x)} +$$

$$\int \frac{\cos x \cdot 5x^4 + x^5 \sin x}{(x^5 \sin x + 5x^4 \cos x)} \sec^2 x dx$$

$$I = -\frac{x}{\cos x (x \sin x + 5 \cos x)} + \tan x + C$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,C,D)

$$f(x) = 3x^2 \text{ and } g(x) = 2x^3 + 1$$

$$\therefore f(x) = g(x) \Rightarrow 2x^3 + 1 - 3x^2$$

$$= (x-1)^2(2x+1) = 0$$

so at (1, 3) curves touches each other

$\therefore$  equation of tangent  $L = 6x - y = 3$

and equation of normal at (1, 3) is

$x + 6y = 19$  which is also passing through (7, 2)

8. Ans (A,B,C,D)

$$\frac{dy}{dx} = -3 \text{ for } f(x) \text{ and } \frac{dy}{dx} = \frac{3}{2} \text{ for } g(x) \text{ at}$$

$$x = -\frac{1}{2}$$

angle between curves is

$$\tan \theta = \frac{\frac{3}{2} + 3}{1 - \frac{9}{2}} \text{ or } \theta = \tan^{-1} \left( \frac{9}{7} \right)$$

$$= \sin^{-1} \left( \frac{9}{\sqrt{130}} \right)$$

Now point of intersection of L and g(x)

$$\Rightarrow 6x - y = 3$$

$$y = 2x^3 + 1 \text{ and point of intersection is } (-2, -15)$$

9. Ans (A,B,C,D)

$$\lim_{x \rightarrow 0} e^{\left(3 + \frac{f(x)}{x^2}\right) \frac{1}{x}} = e^2$$

$$\Rightarrow \lim_{x \rightarrow 0} \left( 2 + \frac{ax^3 + bx^2 + cx + d}{x^2} \right) \frac{1}{x} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \left( \frac{ax^3 + (b+2)x^2 + cx + d}{x^3} \right) = 2$$

$$\Rightarrow a = 2, b = -2, c = 0, d = 0$$

10. Ans (A)

$$2x^3 - 2x^2 - 2(r^2 - 1)x^2 = 0$$

$$2x^3 - 2r^2x^2 = 0$$

$$\Rightarrow \text{Sum of roots} = r^2$$

Sum of roots of all the equations

$$= 1^2 + 2^2 + 3^2 + \dots + 10^2$$

$$= \frac{10 \times 11 \times 21}{6} = 385$$

11. Ans (A)

Apply P-5

$$f(x) = x + \int_0^x e^{-x+t} f(t) dt$$

$$f(x) = x + e^{-x} \int_0^x e^t f(t) dt$$

differentiate w.r.t. x

$$f'(x) = 1 + e^{-x} \cdot e^x f(x) - e^{-x} \int_0^x e^t f(t) dt$$

$$\Rightarrow f'(x) = x + 1$$

$$f(x) = x + \frac{x^2}{2} \quad \boxed{\because f(0) = 0}$$

$$\text{and } \int_1^2 \frac{f''(x)}{x^2} dx = \frac{1}{2}$$

12. Ans (A,D)

Apply P-5

$$f(x) = x + \int_0^x e^{-x+t} f(t) dt$$

$$f(x) = x + e^{-x} \int_0^x e^t f(t) dt$$

differentiate w.r.t. x

$$f'(x) = 1 + e^{-x} \cdot e^x f(x) - e^{-x} \int_0^x e^t f(t) dt$$

$$\Rightarrow f'(x) = x + 1$$

$$f(x) = x + \frac{x^2}{2} \quad \boxed{\because f(0) = 0}$$

$$\text{and } \int_1^2 \frac{f''(x)}{x^2} dx = \frac{1}{2}$$

## PART-3 : MATHEMATICS

### SECTION-II

1. **Ans ( 1.00 )**

$$I = \int e^x \left( \frac{x^2 + 1}{(x^2 + 1)^{3/2}} - \frac{x}{(x^2 + 1)^{3/2}} \right) dx$$

$$\Rightarrow I = \int e^x \left( \frac{1}{\sqrt{x^2 + 1}} + \left( -\frac{x}{(x^2 + 1)^{3/2}} \right) \right) dx$$

$$I = \frac{e^x}{\sqrt{x^2 + 1}} + C$$

$$\therefore f(x) = \frac{1}{\sqrt{x^2 + 1}}$$

Range is (0, 1]

2. **Ans ( 5.00 )**

$$\frac{f'(x)}{f(x)} = \frac{{}^nC_0}{{}^nC_{0x+1}} + \frac{{}^nC_1}{{}^nC_{1x+2}} + \dots + \frac{{}^nC_n}{{}^nC_{nx+n+1}}$$

$$\Rightarrow \frac{f'(0)}{f(0)} = \frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \dots + \frac{{}^nC_n}{n+1}$$

$$f'(0) = (n+1)! \left( \frac{2^{n+1} - 1}{n+1} \right) = n!(2^{n+1} - 1)$$

$$\therefore n! (2^{n+1} - 1) = \frac{3}{2} 7!$$

$$\Rightarrow n = 5$$

3. **Ans ( 6.00 )**

$$\lim_{x \rightarrow 0} \left( 3 - \frac{P(x)}{x} \right) = 27$$

$\Rightarrow P(x)$  has no constant term

$$\text{let } P(x) = ax^4 + bx^3 + cx^2 + dx$$

$$\Rightarrow 3 - d = 27 \Rightarrow d = -24$$

$$P(x) = ax^4 + bx^3 + cx^2 - 24x$$

$$P'(2) = 0, p(1) = -9, p'''(2) = 0$$

$$P'(x) = 4ax^3 + 3bx^2 + 2cx - 24$$

$$P''(x) = 12ax^2 + 6bx + 2c$$

$$P'''(x) = 24ax + 6b$$

$$a + b + c - 24 = -9$$

$$\Rightarrow a + b + c = 15 \quad \dots (1)$$

$$P'(2) = 0$$

$$\Rightarrow 4a(8) + 3b(4) + 2c(2) - 24 = 0$$

$$\Rightarrow 8a + 3b + c = 6 \quad \dots (2)$$

$$P'''(2) = 0$$

$$\Rightarrow 24a(2) + 6(b) = 0$$

$$\Rightarrow 8a + b = 0 \quad \dots (3)$$

Solving (1), (2) & (3)

$$a = 1, b = -8, c = 22$$

$$\Rightarrow P(x) = x^4 - 8x^3 + 22x^2 - 24x$$

$$P'(x) = 4x^3 - 24x^2 + 44x - 24$$

$$= 4[x^3 - 6x^2 + 11x - 6]$$

$$P'(x) = 4[(x-1)(x-2)(x-3)]$$

$$P''(x) = 4[3x^2 - 12x + 11] > 0 \quad \forall x \in [3, 4]$$

$$\Rightarrow P'(x) = 4[(4-1)(4-2)(4-3)]$$

$$= 4[(3)(2)(1)]$$

$$= 24 = 4M$$

$$\Rightarrow M = 6$$

4. **Ans ( 0.41 to 0.42 )**

$$3\sin^{-1} \left( \frac{2x}{1+x^2} \right) + 2\tan^{-1} \left( \frac{2x}{1-x^2} \right) - 4\cos^{-1} \left( \frac{1-x^2}{1+x^2} \right) = \frac{\pi}{4}$$

$$\Rightarrow 6\tan^{-1}x + 4\tan^{-1}x - 8\tan^{-1}x = \frac{\pi}{4}$$

$$\Rightarrow 2\tan^{-1}x = \frac{\pi}{4} \Rightarrow \tan^{-1}x = \frac{\pi}{8}$$

$$\Rightarrow x = \tan \left( \frac{\pi}{8} \right) = \sqrt{2} - 1 = 0.41$$

5. **Ans ( 2.00 )**

$$I_1 = \int_0^{\pi} f(\cos x) dx, I_2 = \int_0^{\pi/2} f(\sin x) dx$$

put  $x = u$  in  $I_1$ , and  $\sin x = u$  in  $I_2$

$$\therefore I_1 = \int_{-1}^1 \frac{f(u)}{\sqrt{1-u^2}} du, I_2 = \int_0^1 \frac{f(u)}{\sqrt{1-u^2}} du$$

$$I_1 = 2 \int_0^1 \frac{f(u)}{\sqrt{1-u^2}} du \Rightarrow I_1 = 2I_2 \quad \therefore \frac{I_1}{I_2} = 2$$

6. **Ans ( 2.11 )**

$f(x)$  is differentiable at  $x = 3$

$\Rightarrow f(x)$  is continuous at  $x = 3$

$$f(3) = \lim_{x \rightarrow 3} f(x)$$

$$\Rightarrow 9p + 3q + 1 = 0 \quad \dots (1)$$

$$f'_+(3) = f'_-(3)$$

$$\Rightarrow 6p + q = 1 \quad \dots (2)$$

$$p = \frac{4}{9} \quad q = \frac{-5}{3}$$